MRC Lecture 6 — Derivation Guide 1

The Convergence Equation Is Unique

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Sources: Methods for Molecular Recognition Computing, Supplementary Note S1; Computational Convergence, Supplementary Note S1 (antibody–antigen Boltzmann selection).

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1. Motivation

Lectures 4 and 5 showed that the Boltzmann distribution governing molecular binding and the softmax attention mechanism in transformers are the same function. That was an observation. This guide turns the observation into a theorem: the softmax/Boltzmann form is the **unique** function satisfying five axioms that any competitive molecular selection process at thermal equilibrium must obey. Once uniqueness is established, the identity between binding and attention is no longer a striking parallel — it is a logical necessity. Two systems that both perform competitive selection at equilibrium *must* implement the same equation, because there is only one equation that can do the job. The argument is due to Luce (1959), who proved it for rational choice without any reference to physics. We restate it in the language of molecular binding, work the decisive calculation (the IIA stress-test) with explicit numbers, build the elimination table, and anchor the result with a worked antibody–antigen example.

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2. The Selection Problem

Every molecular recognition system reduces to one abstract problem. An **agent** (antibody, transcription factor, enzyme, transformer query) has **compatibility scores** $s_1, s_2, \ldots, s_M$ with M candidate **targets** (antigens, DNA sites, substrates, key tokens). We need a function that converts the score vector into a **selection probability distribution** $P_1, P_2, \ldots, P_M$.
We require five properties of this function. Each is a physical statement about competitive binding at thermal equilibrium, not a mathematical convenience. Each eliminates candidate functions. After five, exactly one function remains.

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3. The Five Axioms

Axiom 1 — Positivity. $P_j > 0$ for every candidate j and every score vector. At any nonzero temperature, every reachable binding state has nonzero probability; a function that assigns exactly zero to a reachable state contradicts thermodynamics.
Axiom 2 — Normalization. $\sum_{j=1}^{M} P_j = 1$. The agent binds one target at a time; the probabilities over binding states sum to one.
Together, Axioms 1 and 2 force the structure:

$$P_j = \frac{g(s_j)}{\sum_{l=1}^{M} g(s_l)}, \qquad g(s) > 0$$

for some strictly positive function g. **The entire problem reduces to: what is g?**
Axiom 3 — Unrestricted domain. g must accept any real number as input. Binding energies span the full real line (favorable $\Delta G < 0$, unfavorable $\Delta G > 0$); transformer dot-product scores are unbounded. This eliminates $g(s) = \sqrt{s}$ and $g(s) = \log(s)$ (undefined for negative input).
Axiom 4 — Rank preservation. If $s_A > s_B$ then $P_A > P_B$. A function that scrambles the ordering of scores is injecting noise, not computing selection. This requires g strictly increasing. It eliminates even polynomials ($g(s) = s^2$ maps $+10$ and -10 to the same value), periodic functions (sin rises and falls), and absolute value ($|s|$ cannot distinguish favorable from unfavorable).

Axiom 5 — Independence of Irrelevant Alternatives (IIA). The ratio P_A/P_B depends only on s_A and s_B, not on any other candidate in the pool:

$$\frac{P_A}{P_B} = \frac{g(s_A)}{g(s_B)} = h(s_A - s_B) \quad \text{for some function } h$$

Physically: the binding free energy of a pair is a property of that pair, fixed by their structures and the thermodynamics. Adding a third antigen C to the solution does not change the ΔG of the A pair or the B pair, so it cannot change the A-vs-B odds. **This is the decisive axiom.**

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4. The IIA Stress-Test (The Decisive Calculation)

Axioms 1–4 leave several survivors. The most plausible is the **sigmoid**, $\sigma(s) = 1/(1 + e^{-s})$, which is positive, defined on all of $\mathbb{R}$, and strictly increasing. We test it against IIA with explicit numbers. IIA requires that the ratio $g(s_A)/g(s_B)$ depend only on the difference $s_A - s_B$. We hold the difference fixed at $s_A - s_B = 1$ and vary the absolute scores.

Sigmoid, small scores: $(s_A, s_B) = (1,0)$

$$\sigma(1) = \frac{1}{1 + e^{-1}} = \frac{1}{1.3679} = 0.7311, \qquad \sigma(0) = 0.5000$$

$$\frac{\sigma(1)}{\sigma(0)} = \frac{0.7311}{0.5000} = \boxed{1.46}$$

Sigmoid, large scores: $(s_A, s_B) = (8,7)$

$$\sigma(8) = \frac{1}{1 + e^{-8}} = 0.99966, \qquad \sigma(7) = \frac{1}{1 + e^{-7}} = 0.99909$$

$$\frac{\sigma(8)}{\sigma(7)} = \frac{0.99966}{0.99909} = \boxed{1.00}$$

The sigmoid ratio moved from 1.46 to 1.00 even though the score difference was held fixed at 1. The relative preference between A and B depends on the absolute scores, hence on where the rest of the pool sits. Sigmoid **fails IIA**.

Exponential, both cases

$$\frac{\exp(1)}{\exp(0)} = e^1 = 2.718, \qquad \frac{\exp(8)}{\exp(7)} = e^1 = 2.718$$

The exponential ratio is $\exp(s_A - s_B) = e^1$ in **both** cases — identical, because the normalizer cancels:

$$\frac{\exp(s_A)}{\exp(s_B)} = \exp(s_A - s_B)$$

The ratio depends only on the score difference, never on the absolute scores or on what else is in the pool. The exponential **satisfies IIA exactly.**

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5. The Elimination Table

Collecting every axiom and what it eliminates:

Axiom	Requirement	Eliminates
1. Positivity	$P_j > 0$ for all j	ReLU, simple (non-positive) normalizations
2. Normalization	$\sum_j P_j = 1$	Forces the $g/\sum g$ structure
3. Unrestricted domain	g defined on all of $\mathbb{R}$	$\sqrt{s}$, $\log(s)$
4. Rank preservation	g strictly increasing	s^2 (even polynomials), sin/cos, $\lvert s \rvert$
5. IIA	$P_A/P_B = h(s_A - s_B)$	**sigmoid, and all remaining alternatives**

After five eliminations, one function family survives: the **exponential**, $g(s) = \exp(s/\tau)$.

6. Luce's Uniqueness Theorem

The elimination table is suggestive but not a proof — it rules out specific candidates, not every conceivable function. The proof of uniqueness routes through a functional equation.

Setup. IIA states $g(s_A)/g(s_B) = h(s_A - s_B)$. Setting $s_B = 0$ and writing $g(0) = c$, we get $g(s_A) = c \cdot h(s_A)$. Substituting back, h must satisfy the multiplicative **Cauchy functional equation**:

$$h(a+b) = h(a) \cdot h(b)$$

Solution. The only solutions of $h(a+b) = h(a)h(b)$ that are monotonic (equivalently, measurable — Aczél, *Lectures on Functional Equations*, 1966) are the exponentials:

$$h(s) = \exp(s/\tau)$$

for some constant τ. Combined with rank preservation ($\tau > 0$), this gives the unique selection function:

$$\boxed{P(j) = \frac{\exp(s_j/\tau)}{\sum_l \exp(s_l/\tau)}}$$

This is the **convergence equation**. It is softmax to the machine learning community, the Boltzmann distribution to physicists, the multinomial logit to economists, and Luce's choice rule to psychologists. The form is completely determined by the five axioms; the temperature τ is the only free parameter.

The full rigorous treatment, including the relaxation from differentiability to measurability via Aczél's theorem, is given in Methods for Molecular Recognition Computing, Supplementary Note S1.4. We do not reproduce the measure-theoretic details here; the Cauchy-equation step is the conceptual core.

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7. Worked Example: Three Antibody–Antigen Contacts

To anchor the equation in a molecular setting, consider an antibody paratope with three candidate contacts, each with a compatibility score $s_j = -\Delta G_j/RT$ (negative binding energy in thermal units). Take scores:

$$s_1 = 4.90, \qquad s_2 = 1.65, \qquad s_3 = 0$$

(At $T = 310$ K with $RT = 0.616$ kcal/mol, these correspond to contact energies $\Delta G_1 = -3.02$, $\Delta G_2 = -1.02$, $\Delta G_3 = 0$ kcal/mol — a strong contact, a moderate contact, and a marginal one.)

Step 1: Exponentiate.

$$\exp(4.90) = 134.29, \qquad \exp(1.65) = 5.207, \qquad \exp(0) = 1.000$$

Step 2: Partition function.

$$Z = 134.29 + 5.207 + 1.000 = 140.50$$

Step 3: Probabilities.

$$P_1 = \frac{134.29}{140.50} = 0.955, \qquad P_2 = \frac{5.207}{140.50} = 0.037, \qquad P_3 = \frac{1.000}{140.50} = 0.007$$

$$\boxed{P = (0.955,\ 0.037,\ 0.007)}$$

(Sum = 0.999; the residual is rounding.)

Interpretation. The strongest contact captures 95.5% of the binding probability. A score difference of $s_1 - s_2 = 3.25$ (about 2 kcal/mol) produces a 26-fold preference; a difference of $s_1 - s_3 = 4.90$ (about 3 kcal/mol) produces a 134-fold preference. This steep, exponential sensitivity to small energy differences is the basis of molecular specificity — and it is exactly the behavior of a transformer attention head concentrating its weight on the most compatible key. The same scores fed into softmax give the same distribution; the antibody and the attention head are computing the identical function.

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8. Why This Matters

Uniqueness is what makes the framework deductive rather than suggestive. Because softmax is the *only* function satisfying the five axioms:

- Molecular binding and transformer attention are not analogous — they are the same equation **by necessity**.
- There is no "better than softmax" selection function waiting to be discovered for equilibrium binding; softmax is the right answer, not an approximation to it.
- A neural architecture that respects the physics of molecular recognition **must** compute this equation. This is the bridge to the architecture derivation (WB2) and the training objective (WB3).

A network that appears to outperform softmax on some benchmark is either exploiting non-equilibrium effects (where the axioms, especially IIA, do not hold — see §10) or overfitting an artifact. It is not finding a better selection function, because there isn't one.

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9. Summary

Concept	Result	Significance
Selection problem	scores → probabilities	Common to all recognition systems
Axioms 1–2	$P_j = g(s_j)/\sum g(s_l)$	Forces the structure
Axiom 3	g on all $\mathbb{R}$	Kills $\sqrt{}$, log
Axiom 4	g increasing	Kills s^2, periodic, $\lvert s\rvert$
Axiom 5 (IIA)	ratio $= h(s_A - s_B)$	Kills sigmoid — decisive
IIA stress-test	sigmoid $1.46 \to 1.00$; exp $= e^1$ always	The numerical proof sigmoid fails
Luce's theorem	Cauchy equation $\to g = \exp(s/\tau)$	Uniqueness
Convergence equation	$P(j) = \exp(s_j/\tau)/\sum \exp(s_l/\tau)$	The unique survivor
Worked example	$(0.955, 0.037, 0.007)$	Exponential specificity

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10. Checkpoint Questions

Q1. Repeat the IIA stress-test for the sigmoid at $(s_A, s_B) = (3,2)$ and $(s_A, s_B) = (10,9)$. Confirm the ratio drifts toward 1 as scores grow, while the exponential ratio stays at e^1.
Answer: $\sigma(3)/\sigma(2) = 0.9526/0.8808 = 1.082$; $\sigma(10)/\sigma(9) = 0.99995/0.99988 = 1.0001$. The sigmoid ratio drifts from 1.082 toward 1 — consistent with the $(1,0) \to (8,7)$ result in §4, confirming the ratio depends on absolute scores. The exponential gives $e^1 = 2.718$ in both cases. The sigmoid fails IIA; the exponential satisfies it.
Q2. Show that adding a fourth contact with score $s_4 = 2.0$ to the worked example (§7) changes all the absolute probabilities but leaves the ratio P_1/P_2 unchanged.
Answer: New $Z = 134.29 + 5.207 + 1.000 + \exp(2.0) = 134.29 + 5.207 + 1.000 + 7.389 = 147.89$. New probabilities: $P_1 = 0.908$, $P_2 = 0.035$, $P_3 = 0.0068$, $P_4 = 0.050$. All absolute values dropped (the new candidate took some probability mass). But $P_1/P_2 = 0.908/0.035 = 25.9$, identical to the original $0.955/0.037 = 25.8$ (rounding). This is IIA in action: the relative preference between contacts 1 and 2 is unchanged by the irrelevant fourth alternative.
Q3. A student proposes $g(s) = e^s + 0.01s^2$ as "softmax with a small correction." Which axiom does it violate, and what is the consequence?
Answer: It violates IIA (Axiom 5). The ratio $g(s_A)/g(s_B) = (e^{s_A} + 0.01s_A^2)/(e^{s_B} + 0.01s_B^2)$ does not reduce to a function of $s_A - s_B$ alone — the quadratic terms depend on the absolute scores. Consequence: the predicted preference between two targets would depend on irrelevant third targets in the pool, which contradicts the equilibrium physics (the binding energy of a pair is pair-intrinsic). The "small correction" is not a refinement; it breaks the alignment with the convergence equation. This is the formal reason not to add nonlinearities after the attention softmax.
Q4. The Cauchy functional equation $h(a + b) = h(a)h(b)$ has non-measurable pathological solutions in addition to the exponential. Why does this not weaken the uniqueness claim for molecular recognition?

Answer: The pathological solutions are non-measurable functions constructed using the axiom of choice; they are nowhere continuous and have no physical realizability. Aczél (1966) proved that any *measurable* solution — and a fortiori any monotonic or continuous solution — is exponential. For molecular recognition, the selection probability must vary continuously (in fact smoothly) with the compatibility scores, since scores vary continuously with molecular configuration. Under this physically necessary regularity, the exponential is the unique solution. The pathological solutions are mathematically real but physically excluded. (See Methods MRC S1.4.)

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11. Connection to Future Lectures

- **Lecture 6 WB2:** Substituting the bilinear energy decomposition $s_j = \mathbf{q} \cdot \mathbf{k}_j$ into the convergence equation produces the transformer attention equation directly — turning the unique equation into a prescribed architecture (dual encoders, cross-attention).
- **Lecture 6 WB3:** The maximum-likelihood objective for the convergence equation is InfoNCE — turning the prescribed architecture into a prescribed training objective.
- **Lecture 7:** The five architecture conditions formalize how the convergence equation, plus the physics of molecular recognition, prescribes every component of the SFM. Level 3 alignment is defined: the architecture *is* the governing equation.
- **Lecture 8 onward:** Students build SFMs whose core is this equation. Every domain — given or emergent bilinearity — uses the identical convergence equation; only the encoders and data differ.